

Question number	Answer	Mark
5(a)(i)	The only correct answer is B A is not the correct answer C is not the correct answer D is not the correct answer	1

Question number	Answer	Mark
5(a)(ii)	The only correct answer is B A 0.7 is not the correct answer C 1.3 is not the correct answer D 2.2 is not the correct answer	1

Question number	Answer	Mark
5(a)(iii)	The only correct answer is B A is not the correct answer as lactate production does not decrease C is not the correct answer as lactate production does not decrease D is not the correct answer as oxygen consumption does not increase.	1

Question number	Answer	Additional guidance	Mark
5(b)	A calculation showing the following steps: - calculation of mouse and seal metabolic rate (1) - quotient calculated (1)	$20/10 = 2 \text{ cm}^3\text{g}^{-1}\text{h}^{-1}$ $18200/5000000 = 0.0036 \text{ cm}^3\text{g}^{-1}\text{h}^{-1}(0.00364)$ $2/0.0036 = 555.56/556/549.45/549/549.5$ Correct answer gains all 2 marks	2

Question number	Answer	Additional guidance	Mark
5(c)	An answer that includes the following points: Indicative content • No direct relationship between body mass and lactate production • The larger the size of the crocodile the lower the rate of lactate production • Highest rate of lactate production in 20kg crocodiles • Smallest crocodiles have same rate of lactate production as 50kg crocodiles • Comment on the link between aerobic and anaerobic respiration Description of named responses • As crocodiles were exercising vigorously the aerobic pathway cannot keep up. • ATP generated anaerobically in glycolysis using glycogen stores in the muscle. • Energy used to allow the crocodile to move very rapidly, if only for a short time. Chemical nature of responses • 1 mol of glucose from glycogen generates 3 mols of ATP and 2 mols of lactate • Detailed description of anaerobic pathway in glycolysis • Detailed description of aerobic pathways Detailed mechanism of responses Level 1: Description of (a minimum of 2 specific) results from experimental data. Level 2: All of level 1 plus reference to chemical nature of anaerobic or aerobic respiration Level 3: All of level 2 plus detailed mechanism for anaerobic and aerobic respiration		6

